ConFoo.CA DEVELOPER CONFERENCE

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17 Things Developers Should Know About Open Source Databases

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Founder, Percona

Who Are you?

More Developer?

More OPS?

What Database Technologies do you run

- MySQL/MariaDB
- PostgreSQL
- MongoDB
- Redis/Valkey

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Who Manages Databases for you ?

- Your Team
- Separate team in your Organization
- “Fully Managed” Cloud Service
- AI fully drives my database

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Programming Language

What Programming Languages does your team use?

Devs vs Ops

Is there Separation between Devs and Ops in your Organization?

Any Tension?

Any Tension Between Devs and Ops?

Devs vs Ops

DevOps suppose to have solved it but tension is still common between Devs and Ops

Especially with Databases which are often special snowflake

Especially with larger organizations

Large Organizations

Ops vs Ops have conflict too

Devs vs Ops Conflict

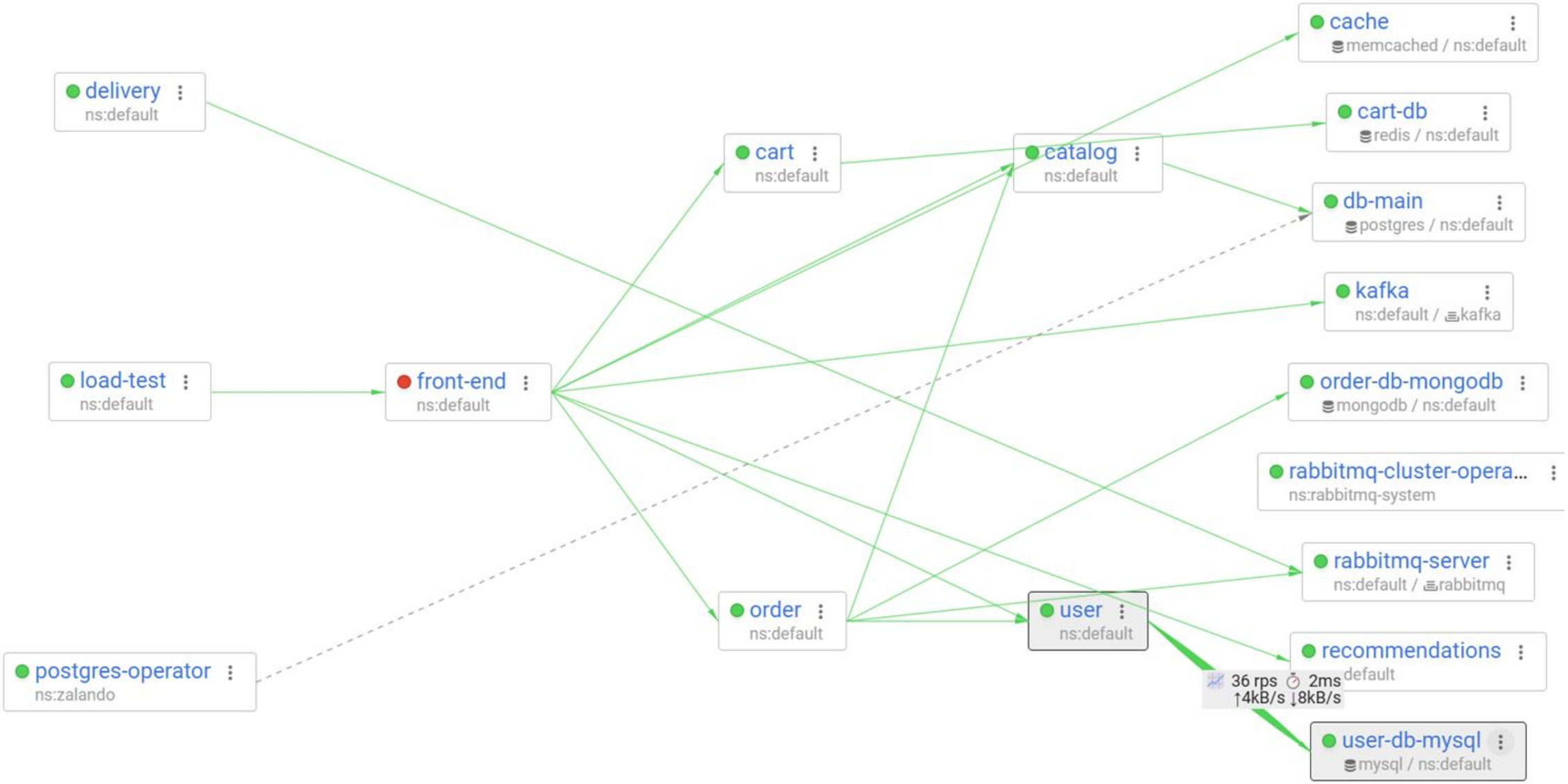
Devs	Ops
• Why is this stupid database always the problem? • Why can't it just work and work fast?	• Why do not learn schema design? • Why do not you write optimized queries? • Why do not you think about capacity planning?

Database Responsibility

Shared Responsibility for Ultimate Success

Top Recommendations for Developers

Is it even Database ?



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Ways to run a database

- “What Database ?”
- Fully Managed (Cloud)
- Cloud Native (Kubernetes)
- VMs or Bare Metal

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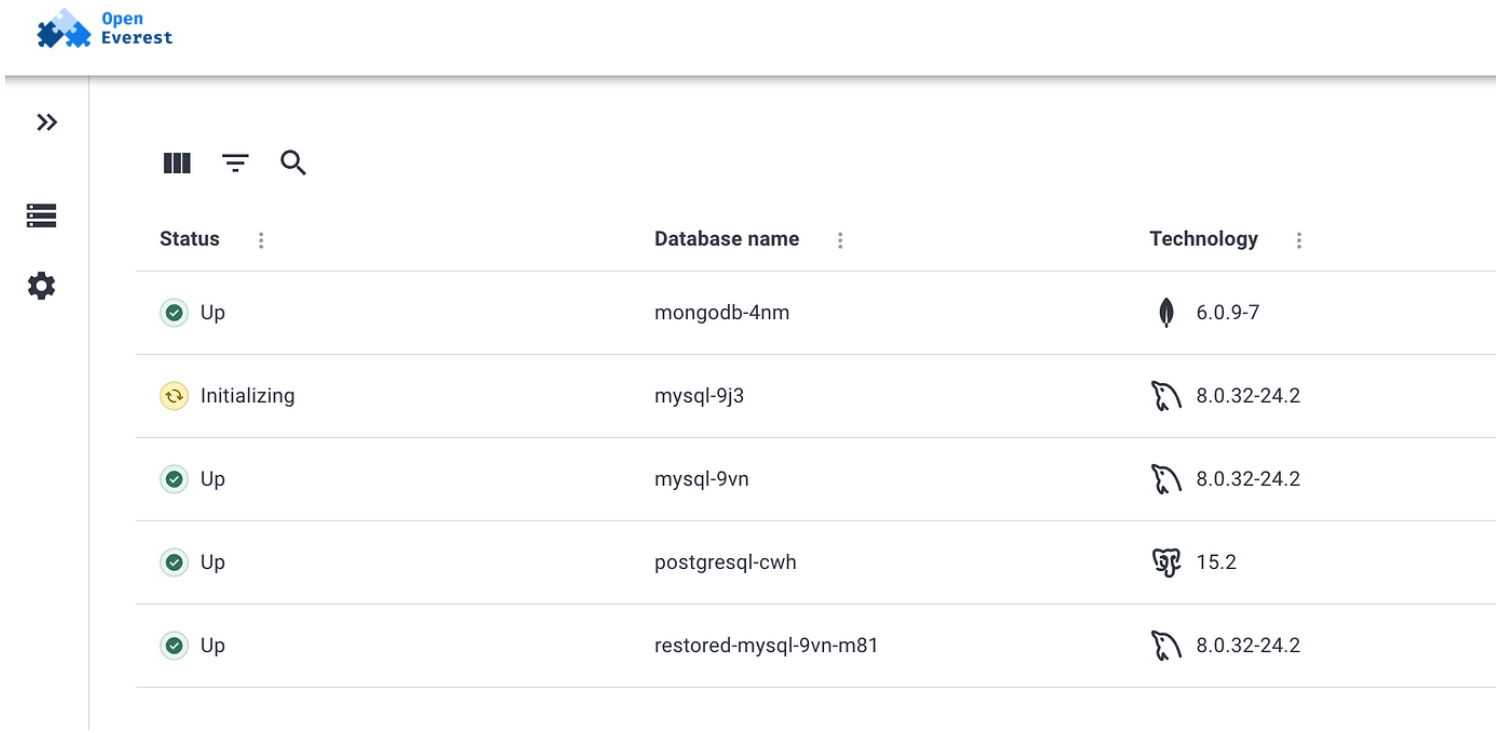

Cloud Native Gets Easier with OpenEverest

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Learn Database Basics

You can’t build a great database powered applications if you do not understand how databases work

Schema Design

Power of the Database Language

How Database Executes the Query

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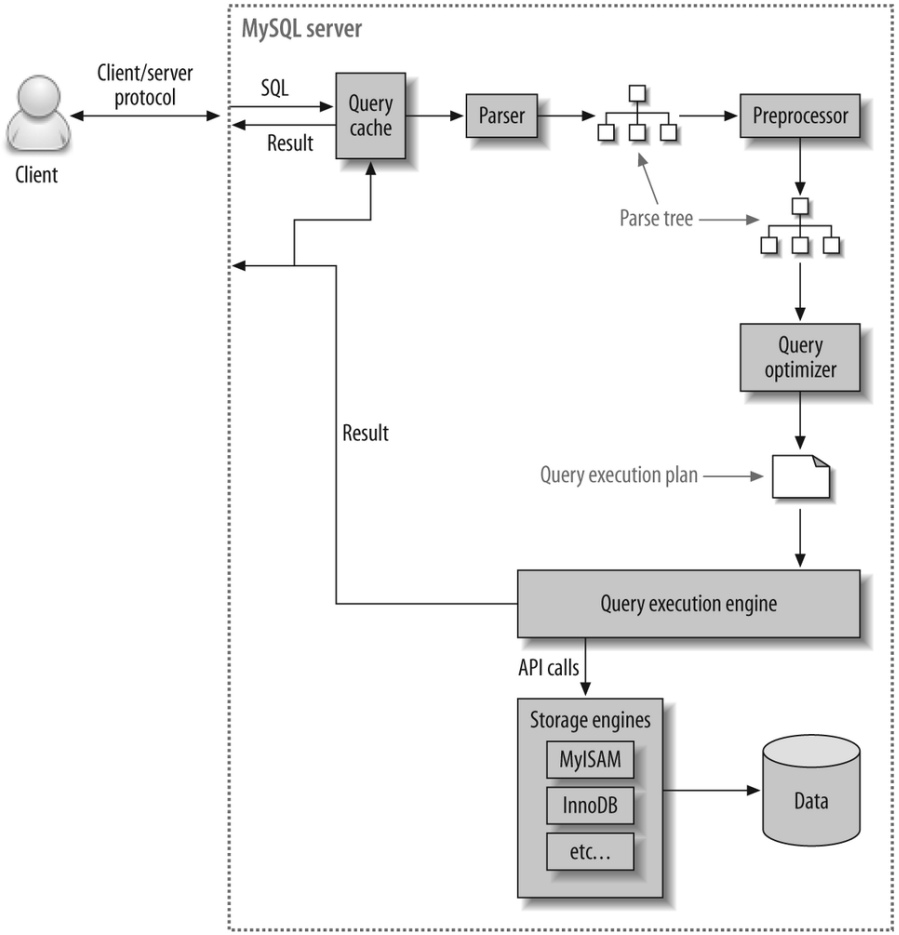
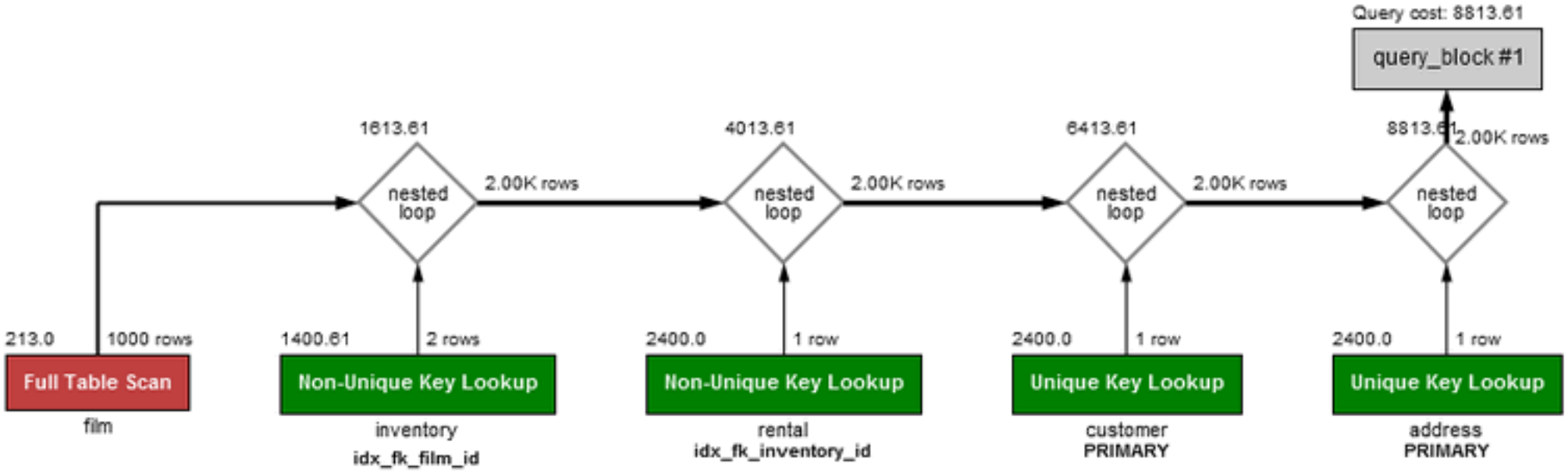
Query Execution Diagram

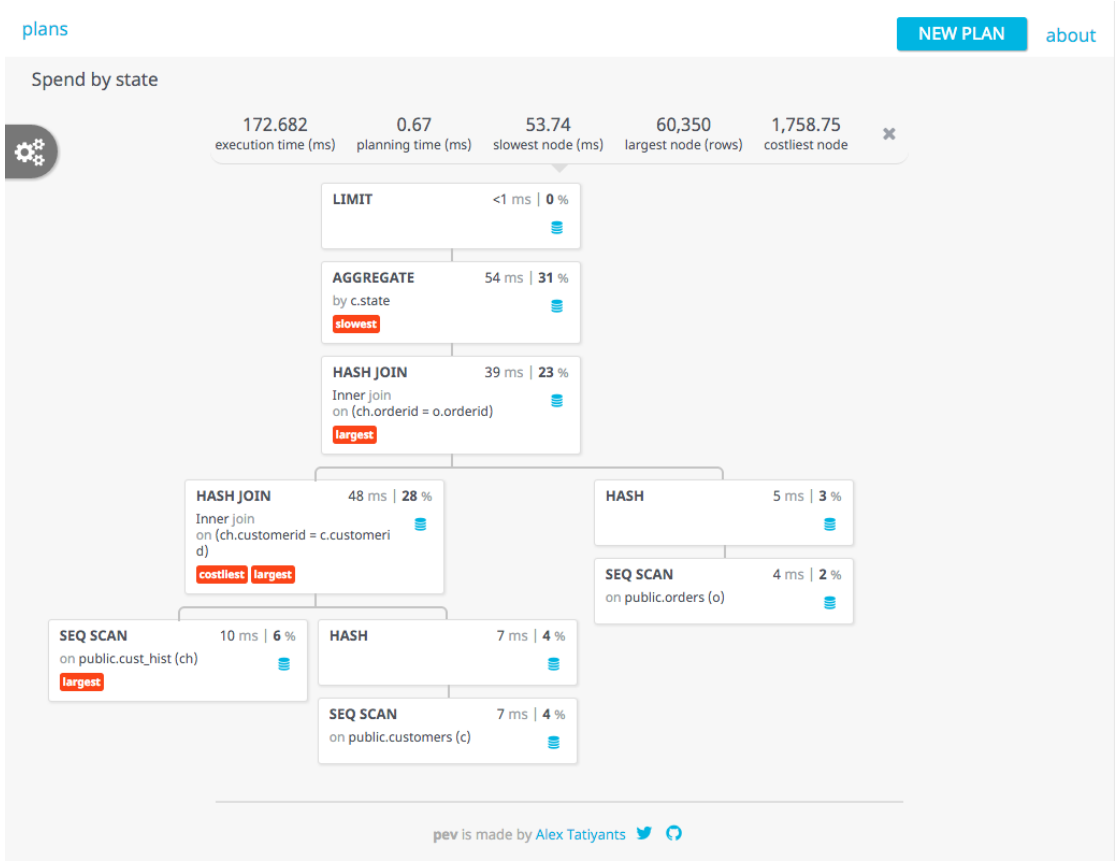


EXPLAIN



https://dev.mysql.com/doc/refman/8.0/en/execution-plan-information.html

Visualizing PostgreSQL Plan with PEV



http://tatiyants.com/postgres-query-plan-visualization/

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Which Queries are Causing the Load

Environment

mytest	99.99%
n/a	0.01%

Database

n/a	99.99%
pmm-managed	<0.01
postgres	<0.01

Schema See all

tpcc1	20.44%
tpcc5	17.36%
tpcc3	17.32%
tpcc2	17.31%
tpcc4	17.29%

Node Name

mysql2	41.31%

#	Query	Load			Query Count		
	TOTAL		1.18	100 %	953.18	41.18m	100 %
1	select c from sbtest1 where id=?		0.35	29.25 %	497.32	21.48m	52.17 %
2	update warehouse1 set w_ytd = w_...		0.14	11.99 %	6.24	269.36k	0.65 %
3	select d_next_o_id, d_tax from distr...		0.10	8.34 %	6.23	269.28k	0.65 %
4	update district1 set d_ytd = d_ytd + ...		0.09	7.55 %	6.24	269.36k	0.65 %
5	commit		0.08	6.86 %	20.48	884.79k	2.15 %
6	select i_price, i_name, i_data from i...		0.06	5.07 %	62.32	2.69m	6.54 %
7	insert into new_orders1 (no_o_id, n...		0.05	4.3 %	6.23	269.28k	0.65 %
8	select count(distinct (s_i_id)) from o...		0.04	3.04 %	0.62	26.62k	0.06 %
9	select o_id from orders1 o, (select ...		0.04	2.97 %	6.21	268.28k	0.65 %

Why Are they Causing this Load

SELECT sbtest

737F39F04B198EF6

Metrics

Query first seen: Aug 3, 2017 1:55 PM ••• Last seen: Today at 9:46 AM

Metrics	Rate/Sec	Sum	Per Query Stats
Query Count	104.05 (per sec)	374.58 k 4.27% of total	
Query Time	19.00 load	18:59:56 29.73% of total	183.66 ms avg
Lock Time	0.11 (avg load)	0:06:42 1.35% of total 0.61% of query time	1.13 ms avg
Innodb IO Read Wait	0.61 (avg load)	0:36:44 9.10% of total 3.38% of query time	6.20 ms avg
Innodb Read Ops	52.35 (per sec)	188.45 k 7.62% of total	0.00 avg
Innodb Read Bytes	857.64 KB (per sec)	3.09 GB 7.62% of total 16.38 KB avg io size	8.22 KB avg
Innodb Distinct Pages	-	-	4.69 avg
Rows Sent	10.41 k (per sec)	37.46 m 30.52% of total	100.00 avg
Bytes Sent	1.30 MB (per sec)	4.67 GB 30.78% of total 124.71 Bytes bytes/row	12.47 KB avg
Rows Examined	1.14 m (per sec)	4.11 b 39.17% of total 109.79 per row sent	10.47 k avg
External Sorts (Filesort)	104.05 (per sec)	374.58 k 49.93% of total 100.00% of queries	-
Full Table Scans	0.01 (per sec)	40.00 0.17% of total 0.01% of queries	-
Queries Requiring Tmp Table In Memory	104.05 (per sec)	374.58 k 95.17% of total 100.00% of queries	-

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How to Improve their Performance

Example

```
SELECT DISTINCT c
FROM sbtest1
WHERE id
    BETWEEN 5559
        AND 5658
ORDER BY  c
```

CREATE

```
CREATE TABLE `sbtest1` (
  `id` int(10) unsigned NOT NULL AUTO_INCREMENT,
  `k` int(10) unsigned NOT NULL DEFAULT '0',
  `c` char(120) NOT NULL DEFAULT '',
  `pad` char(60) NOT NULL DEFAULT '',
  PRIMARY KEY (`id`),
  KEY `k_1` (`k`)
) ENGINE=InnoDB AUTO_INCREMENT=100000001 DEFAULT
```

JSON

Expand All


```
-{
    "query_block": -{
        "select_id": 1,
        "cost_info": +{...},
        "ordering_operation": -{
            "using_filesort": false,
            "duplicates_removal": -{
                "using_temporary_table": true,
                "using_filesort": true,
                "cost_info": +{...},
                "table": +{...}
```

How are Queries Executed?

Single Threaded

Single Node

Distributed

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Indexes

Indexes are Must

Indexes are Expensive

Get proficient talking to AI

What should I do if MySQL is not using the index I think it should use

If MySQL is not using the index you expect, it can lead to slow query performance. Here are steps you can take to diagnose and resolve the issue:

---

1. Verify the Index Exists

- Ensure the index you think should be used actually exists on the table.
- Run:

```sql
SHOW INDEX FROM your_table_name;
```

- Check if the index is listed and matches the columns in your query.

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Capacity Planning

No Database can handle “unlimited scale”

Scalability is very application dependent

Trust Measurements more than Promises

Can be done or can be done Efficiently?

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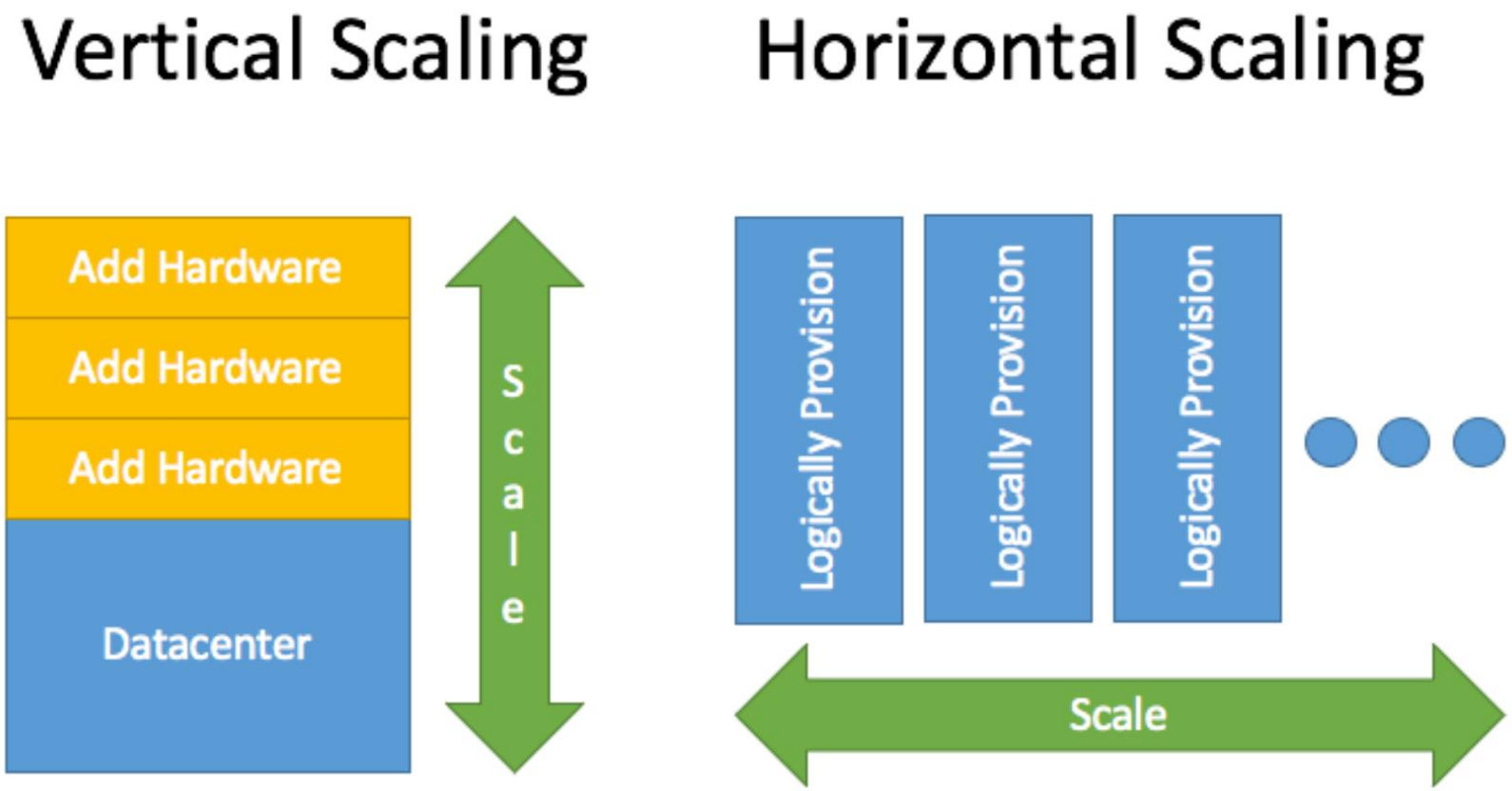
Vertical and Horizontal Scaling



Scalable != Efficient

The Systems which are “scalable” can be less efficient

Hadoop, Cassandra, TiDB are great examples

By only the wrong thing you can get in trouble

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Throughput != Latency

If I tell you system can do 100.000 queries/sec would you say it is fast?

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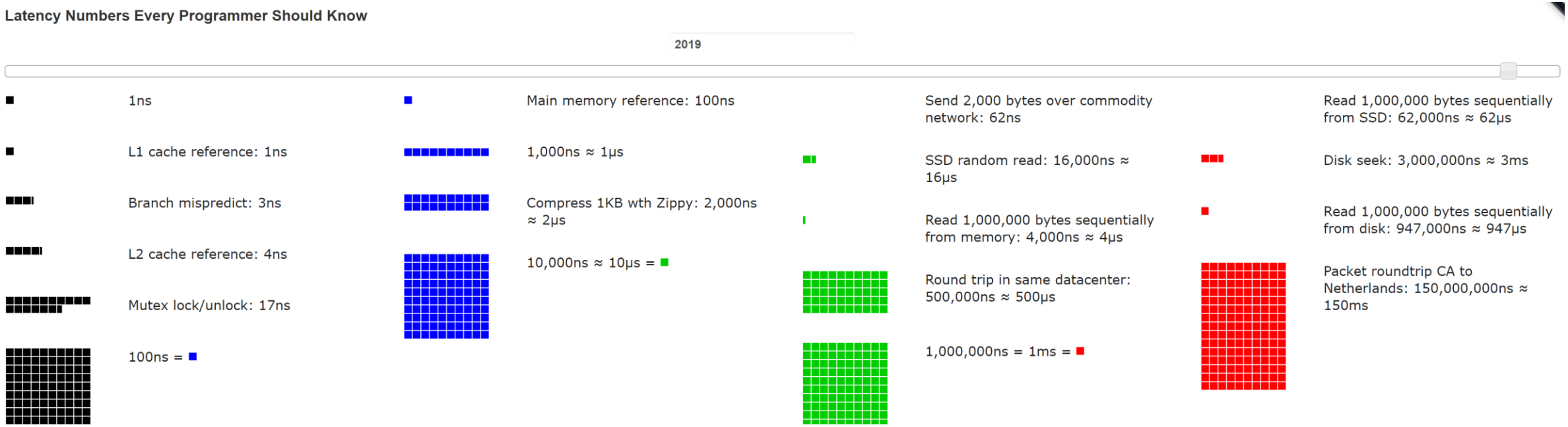
Latency Numbers to Know



Very cool Interactive Diagram
https://colin-scott.github.io/personal_website/research/interactive_latency.html

Speed of Light Limitations

High Availability Design Choices

You want instant durable replication over wide geography or Performance?

Understanding Difference between High Availability and Disaster Recovery protocols

Network Bandwidth is not the same as Latency

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Wide Area Networks

Tend not to be as stable as Local Area Networks (LAN)

Expect increased Jitter

Expect Short term unavailability

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Also Understand

- Connections to the database are expensive
- Especially if doing TLS Handshake
- Query Latency Tends to Add Up
- Especially on real network and not your laptop

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ORM (Object-Relational-Mapping)

- Allows Developers to query the database without need to understand SQL
- Can create SQL which is very inefficient
- Learn SQL Generation “Hints”, Learn JPQL/HQL advanced features
- Be ready to manually write SQL if there is no other choice

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Do Not Leave Transactions Open

- Open Connection is rather inexpensive
- Transaction open for Long Time can get very expensive (even if it performed no writes)
- Isolation Mode Matters
- SET AUTOCOMMIT=0 - Any SELECT query will Open Transaction
- COMMIT/ROLLBACK closes transaction

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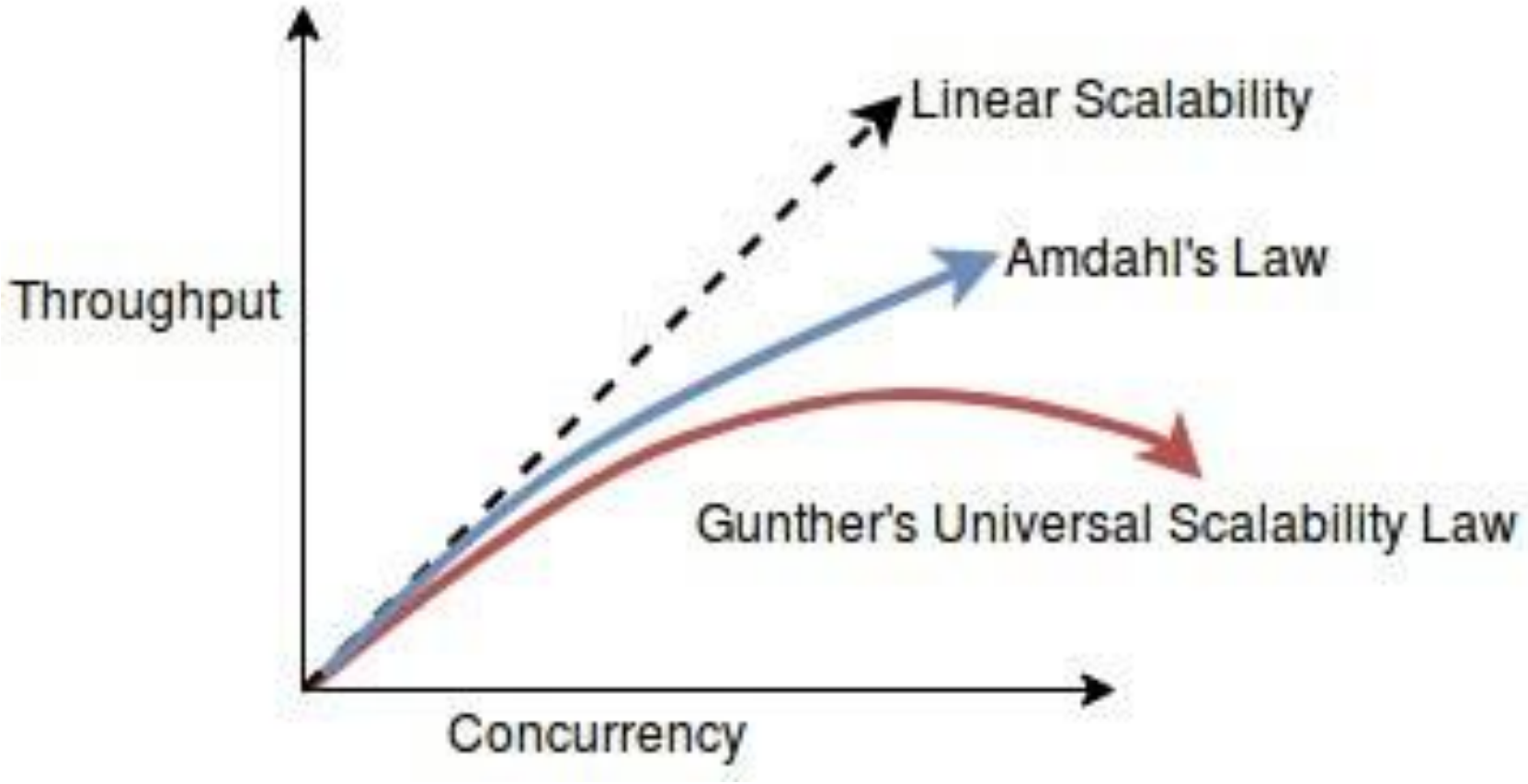
Understanding Optimal Concurrency



Embrace Queueing

Request Queueing is Normal

With requests coming at “Random Arrivals” some queueing will happen with any system scale

Should not happen to often or for very long

Queueing is “Cheaper” Close to the User

Benefits of Connection Pooling

Avoiding Connection Overhead, especially TLS/SSL

Avoiding using Excessive Number of Database Connections

Multiplexing/Load Management

Configuring Connection Pool

- **Default and Maximum Connection Pool Size**
- **Scaling Parameters**
- **Combined Connection Pool Max Size should be smaller than number of connections database can support**
- **Waiting for free connection to become available is OK**

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Law of Gravity

Shitty Application at scale will bring down any Database

Scale Matters

Developing and Testing with Toy Database is risky

Queries Do not slow down linearly

The slowest query may slow down most rapidly

Memory or Disk

- Data Accessed in memory is much faster than on disk
- It is true even with modern SSDs
- SSD accesses data in large blocks, memory does not
- Fitting data in Working Set

Upgrading to the new Software/Hardware is not always faster

Test it out

Defaults Change are often to blame

Newer is not Always Faster

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Major Database Upgrades often require application changes

Having Conversation on Application Lifecycle is a key

Upgrades are needed but not seamless

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Operational Overhead

Operations Take Time, Cost Money, Cause Overhead

10TB Database Backup?

Adding The Index to Large Table?

Distributed Systems

- 10x+ More Complicated
- Better High Availability
- Many Failure Scenarios
- Test how application performs

Risks of Automation

Automation is a Must

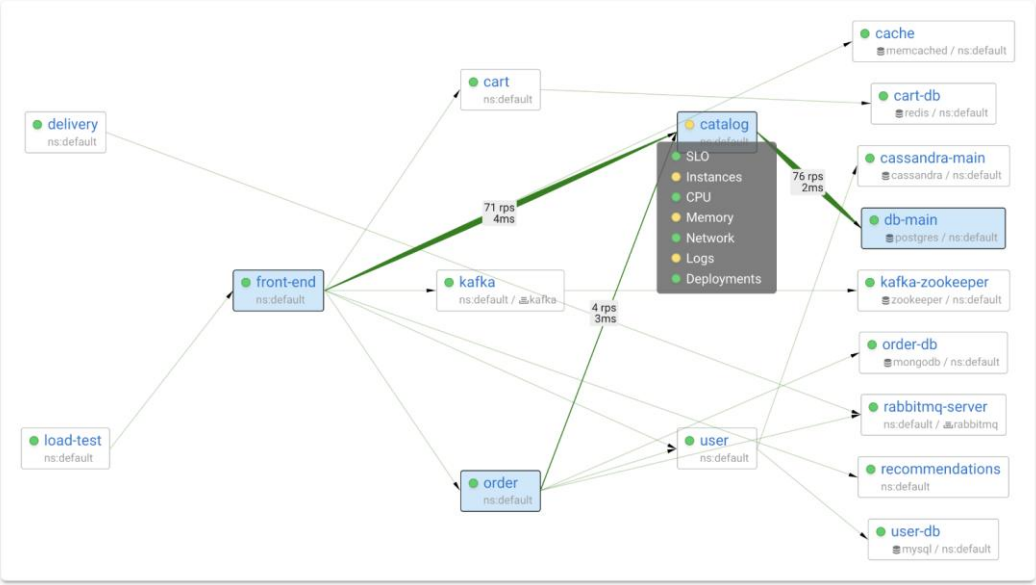
Mistakes can destroy database at scale

Security

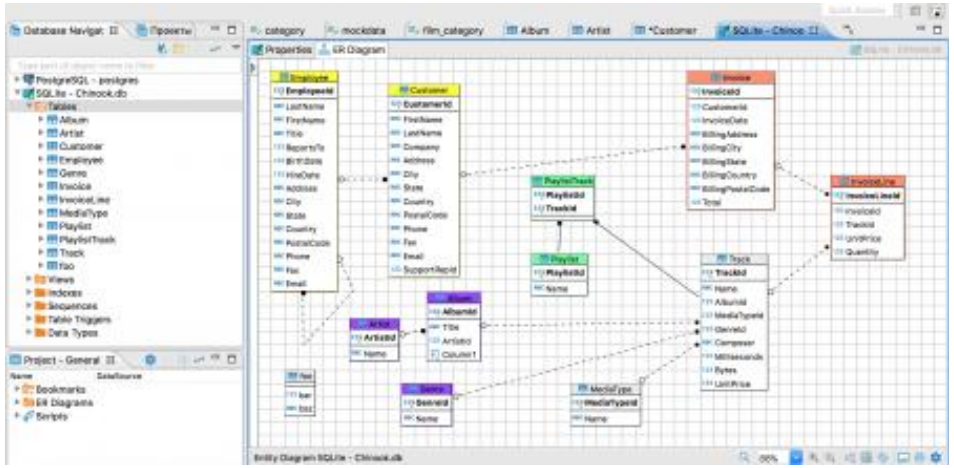
Database is where the most sensitive data tends to live

Shared Devs and Ops Responsibility

Many Wonderful tools



- **DBeaver**
- **Releem**
- **Coroot**

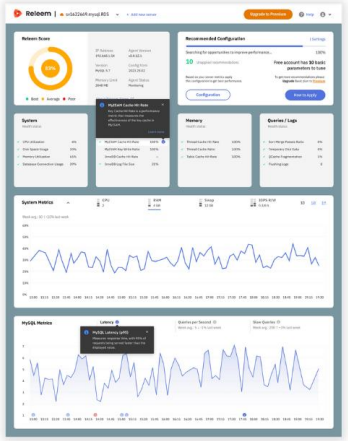


Releem

Easy-to-Use

MySQL Performance Tuning and Monitoring Tool

- Simple Installation
- Safe
- Automatic



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Using AI

BUILD

MANAGE

INTERFACE

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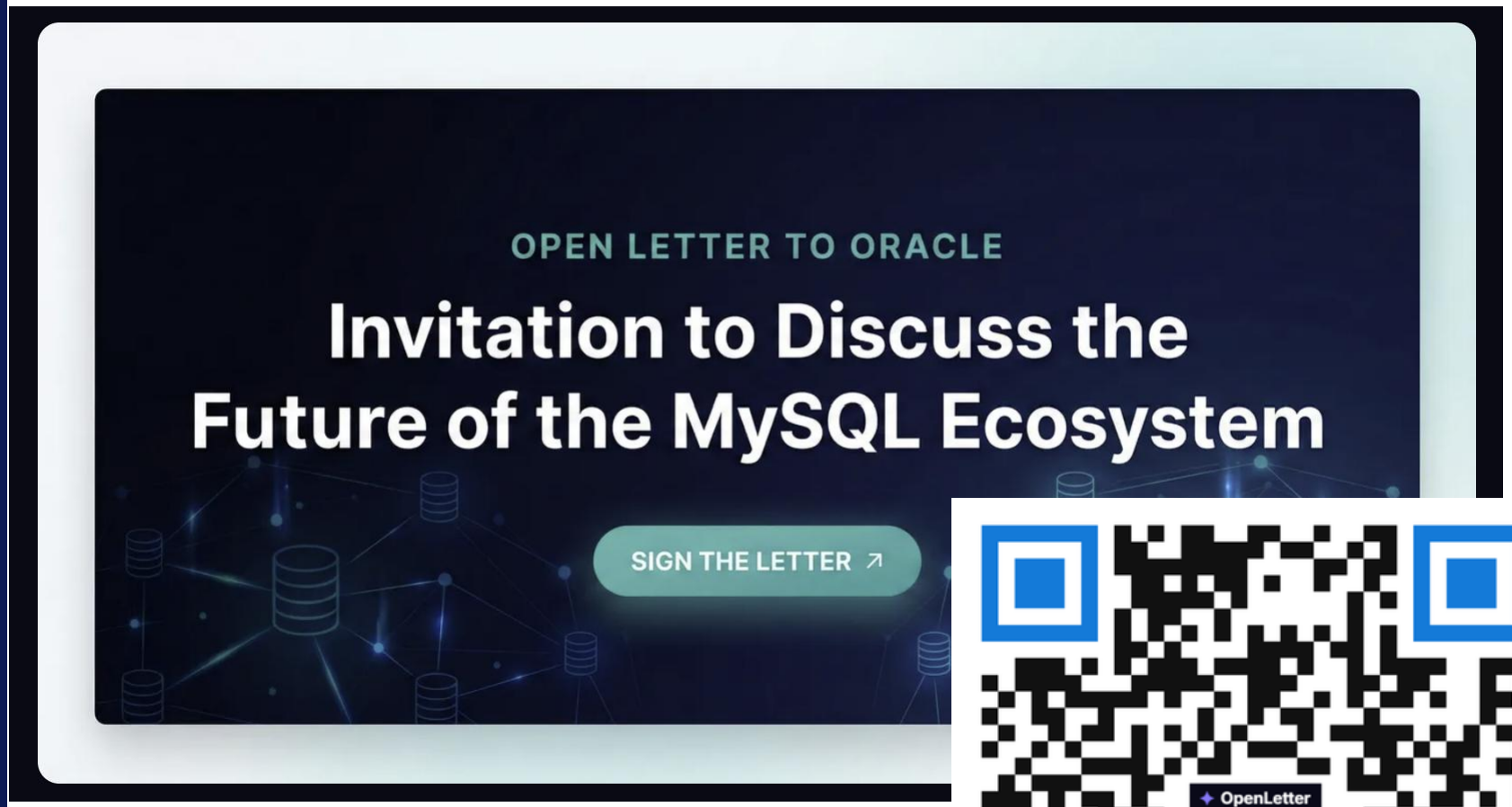
What Else

What Would you Add ?

Public Service Messages

Do you think MySQL Users would benefit from Independent Association having their back ?



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Computer History Museum,
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